

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

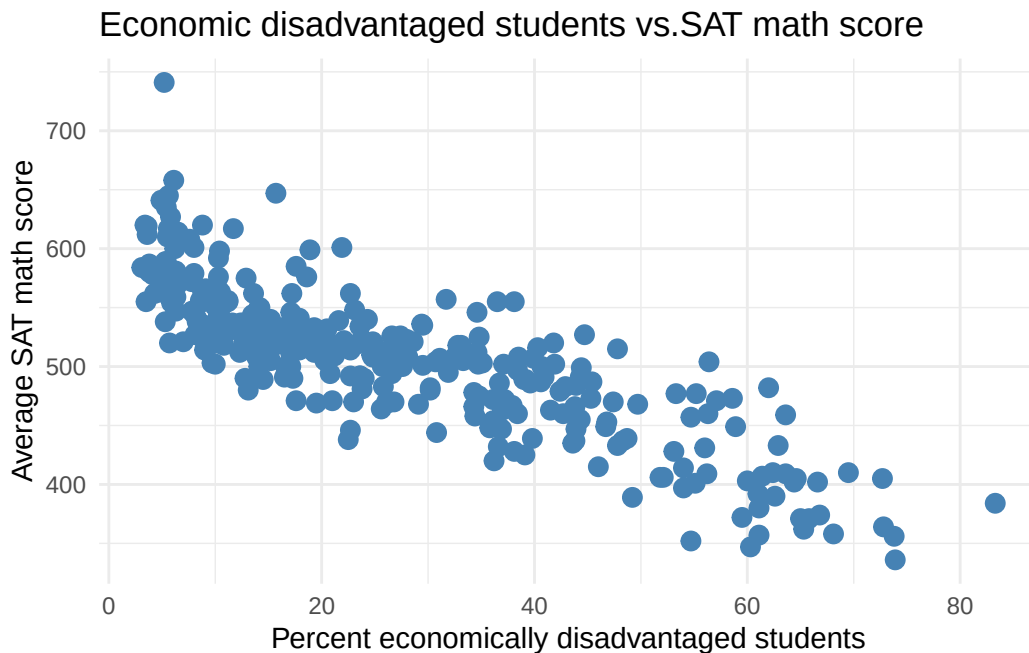
# Load the MA_schools dataset
data(MA_schools)
```

Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
ggplot(MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +  
  geom_point(size = 3, color = "steelblue") +  
  labs(title = "Economic disadvantaged students vs.SAT math score",  
        x = "Percent economically disadvantaged students", y = "Average SAT math score") +  
  theme_minimal()
```



Answer: Based on the scatterplot, there is a negative relationship between the percentage of economically disadvantaged students and average SAT math score. Because while the percentage of economically disadvantaged students increases, the average SAT math score tends to decrease.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)

summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
sat_effects <- coef(model1)

sat_effects
```

(Intercept)	perc_disadvan
581.281054	-2.779787

Intercept: 581.281054

Slope: -2.779787

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: The intercept (581.28) represents the predicted average SAT math score for a school when economically disadvantaged students = 0%. The slope (-2.78) means that for every increase in economically disadvantaged students, the average SAT math score is expected to decrease by about 2.78 points, on average. That means the average SAT math has a negative relationship with economically disadvantaged students. It means that schools with a higher proportion of economically disadvantaged students tend to have lower average SAT math scores, showing a negative relationship between economic disadvantage and academic performance in SAT math.

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
predict(model1, newdata = data.frame(perc_disadvan = 50))
```

```
      1  
442.2917
```

Predicted average SAT math score: Expected the average SAT math score for a school where 50% of students are economically disadvantaged is about 442.29

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
predict(model1, newdata = data.frame(perc_disadvan = 50), interval = "prediction")
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)

summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-81.342 -20.802  -3.872  18.211 167.251
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   588.1898     7.6067  77.325  <2e-16 ***
perc_disadvan  -2.7772     0.1063 -26.120  <2e-16 ***
sizemedium    -11.9134     7.5346  -1.581    0.115
sizelarge     -6.3615     6.9231  -0.919    0.359
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
summary(model1)$r.squared
```

```
[1] 0.6961948
```

```
summary(model2)$r.squared
```

```
[1] 0.6987587
```

Answer: model 2 explains slightly more variance in average SAT math scores compared to model1, because the R-squared of model is a bit higher than model1, with 0.699 and 0.696 respectively. But the improvement is very small (just 0.002), suggesting that adding `size` does not substantially increase the model's explanatory power

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
anova(model1, model2)
```

Analysis of Variance Table

Model 1: `average_sat_math ~ perc_disadvan`

Model 2: `average_sat_math ~ perc_disadvan + size`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	330	371191				
2	328	368058	2	3132.6	1.3958	0.2491

Conclusion: The `p_value` = 0.25, which is greater than 0.05. It means that adding `size` variable does not significantly improve the model. In other words, the multiple regression model like model2 does not perform significantly better than the simple model in explaining average SAT math scores